

ME 600 : Research Methodology

Course Code : ME 600

Course Name : Research Methodology

L-P-T-C: 1-0-0-1

Intended for: MTech(R), PhD

Prerequisites: None

Mutual Exclusion: Research Methodology courses running in various schools

Approval : 57th BoA

Course Contents

- **Overview of Research:** Purpose of research, selection of a research problem, literature review and research gap, various research methodologies, overview of logical reasonings, the scientific method, hypothesis forming (3 Hours)
- **Theoretical Research:** Importance of theory in research, introduction to physical and mathematical modeling, methods of theoretical research (1 Hour)
- **Experiments in Research:** Introduction to experimental research, design of experiments, measurement techniques and instrumentation, uncertainty analysis, best practices in experiments, and ethical considerations in experimental research (2 Hours)
- **Numerical Simulations in Research:** Importance of computational research, advances and applications of computational techniques in various fields, common numerical methods and tools (2 Hours)
- **Scientific Communication:** Elements of scientific communication, introduction to LaTeX, best practices in writing reports, theses and journal articles (6 Hours)

Laboratory/practical/tutorial Modules:

A technical report writing with LaTeX will be part of the course in which the student needs to submit a report based on one's own research work.

Text books:

1. Soumitro Banerjee, **Research Methodology for Natural Sciences**, IISc Press, 2022
2. Wayne Booth, Gregory Colomb, Joseph Williams, Joseph Bizup, William Fitzgerald, **The Craft of Research**, 4th Edition, The University of Chicago Press, 2016

References:

1. Kate L Turabian, **A Manual for Writers of Research Papers, Theses and Dissertations**, 9th Edition, The University of Chicago Press, 2018
2. John W. Creswell, J, David Creswell, **Research Design: Qualitative, Quantitative, and Mixed Methods Approaches**, 6th Edition, SAGE Publications, 2022